BloodLink

رابط الحياة

Project Documentation

1. Project Planning & Management
2. Stakeholder Analysis
3. Database Design
4. UI/UX Design

Program	DEPI (Digital Egypt Pioneers Initiative), 2026
Track	Software Development - React Frontend Web Developer
Team	Ibrahim Hassan Anwar (Team Leader) Khaled Atef Salah Mohammed Eslam Mohammed Hager Waleed Shaker Zeyad Mohammed Mohammed Marawan Hesham Tolba
Repository	github.com/Marnie0/bloodlink
Date	31 October 2026

Phase 1: Project Planning & Management

1. Project Proposal

Overview

BloodLink is a community website that connects people in Egypt who need blood with suitable donors nearby. Every request is backed by an official hospital paper. Donor and patient always meet at the hospital, and no money is involved.

Problem

- Blood requests are spread through Facebook and WhatsApp posts that reach people by chance.
- Nobody can tell if a post is real or still needed.
- Most people who see a post have the wrong blood type or live too far away.
- Contact details are posted publicly, which invites scams and requests for money.

Solution

Step	What happens
Donor joins	Uploads an official hospital certificate showing their blood type
Patient asks	Uploads an official hospital paper proving the patient needs blood
Paper check	Partner hospital paper: confirmed by the hospital, badge **Verified**. Other official hospital: checked by AI, badge **Paper looks valid**.
Matching	Compatible, available donors closest to the hospital are notified first, and the area grows in waves
Accept	The donor accepts. Phone numbers are shared only then.
Meet	At the hospital. The hospital decides medically. Partner hospitals scan the donor's QR code to record the donation.

Objectives

1. Send each request only to donors who can actually help: compatible blood type, available, and nearby.
2. Show how much each request and donor can be trusted, using hospital papers and clear badges.
3. Protect users: phone numbers stay hidden until a donor accepts, meetings happen only at hospitals, and money requests are banned.
4. Deliver a working, deployed Arabic website that works well on phones and computers, by 30 Nov 2026.

Scope

In scope (core)	Out of scope
Accounts with 4 roles: donor, requester, partner hospital, admin	Examining donors or making any medical decision
Certificate and hospital paper upload	Guaranteeing a paper is genuine
AI paper check and partner-hospital confirmation	Payments of any kind

In scope (core)	Out of scope
Matching by blood type, distance and trust level	SMS or WhatsApp Business API
Web push notifications in waves	Mobile apps (a possible future improvement; this project is a website only)
Accept or decline, with phone numbers hidden until acceptance	
QR check-in at partner hospitals; self-confirmation elsewhere	
Live request progress	
Admin: paper review, reports, bans, key numbers	
Arabic RTL interface that works on phones	

Planned extras (built only after checkpoint 2): an English version, a map view, a WhatsApp share card, donation history and charts.

Main deliverables

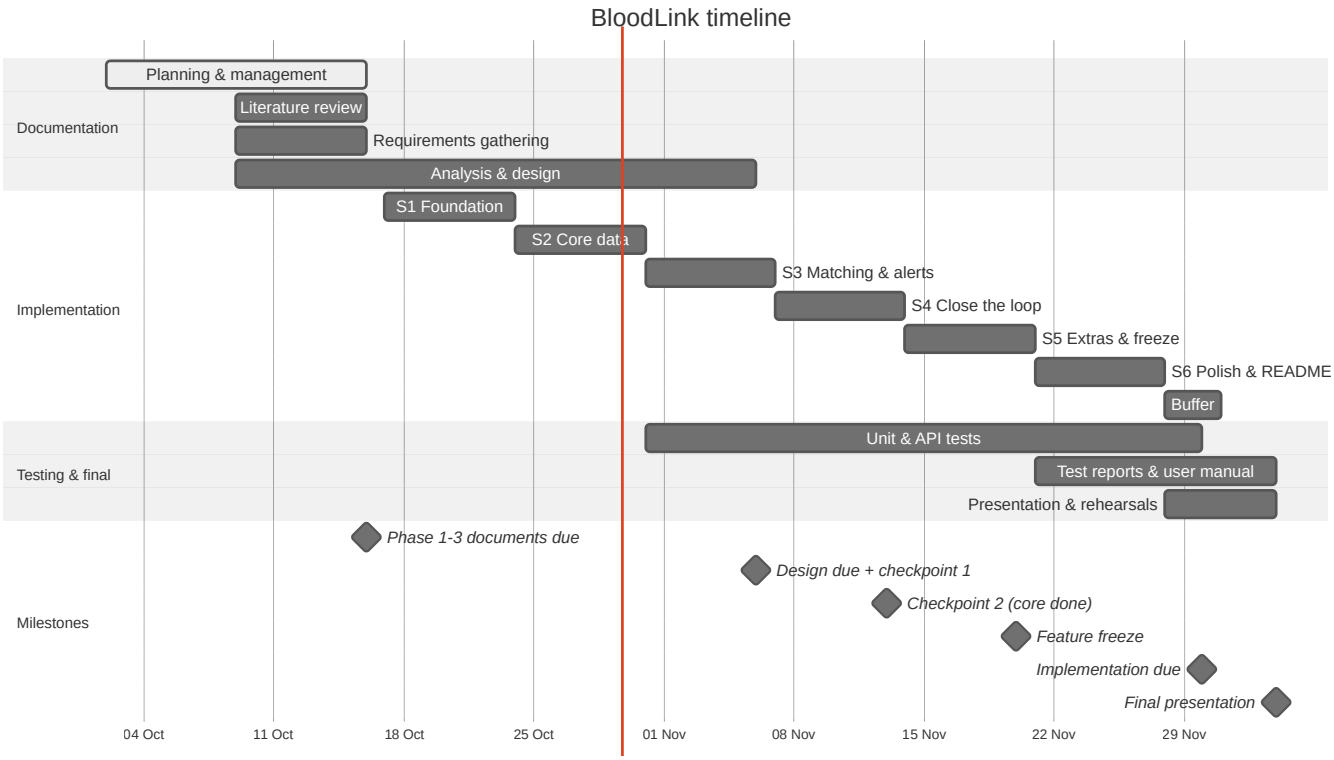
- Deployed website with demo data
- Source code on GitHub, with a README and API documentation
- Phase documents (planning, requirements, design, testing)
- Presentation, demo video and project infographic

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2. Project Plan

Timeline (Gantt chart)

The plan follows the DEPI deadlines. Implementation runs in six one-week sprints, Saturday to Friday, with a three-day buffer before the 30 Nov deadline.



Sprints

Sprint	Dates	Goal	Done when
S1 Foundation	17 to 23 Oct	Repo, Docker, deploy, data models, login with roles, Arabic UI kit	The website is deployed and a user can log in with a role
S2 Core data	24 to 30 Oct	Donor signup and certificate upload, request form and paper upload, partner confirmation	A request becomes active after the hospital confirms it
S3 Matching & alerts	31 Oct to 6 Nov	Matching with tests, AI paper check, push notifications, accept or decline, live progress	**Checkpoint 1:** the whole flow works roughly
S4 Close the loop	7 to 13 Nov	QR check-in at partner hospitals, self-confirmation elsewhere, waves, admin tools	**Checkpoint 2:** all core features are done
S5 Extras & freeze	14 to 20 Nov	Planned extras (if checkpoint 2 passed), security, bug fixes	**Feature freeze**
S6 Polish	21 to 27 Nov	Bug fixes only, README, API docs, demo data	Ready to submit

Milestones and deliverables

Date	Milestone	Deliverable
16 Oct	Phases 1 to 3 submitted	Planning, literature review and requirements documents
23 Oct	Skeleton live	Deployed website with login
6 Nov	Design submitted, checkpoint 1	Diagrams, database design, wireframes, the end-to-end flow working roughly
13 Nov	Checkpoint 2	All core features working
20 Nov	Feature freeze	No new features after this date
30 Nov	Implementation submitted	Source code, README, deployed link

Date	Milestone	Deliverable
4 Dec	Final presentation	Test reports, user manual, slides, demo video, infographic

Resource allocation

Time (worst case). Every member works about 10 hours a week for 8 weeks, from 9 Oct to 4 Dec.

	Hours
Capacity: 6 members × 10 h × 8 weeks	480
Weekly meetings (1 h each per week)	−48
One member loses a full week	−10
Available	**422**
Core development (estimates × 1.5 safety factor)	167
Planned extras (× 1.5)	59
Integration and testing	40
Documentation for all phases (× 1.5)	60
Research, presentation, video, rehearsals (× 1.5)	30
Planned	**356**
Spare	**≈ 66**

The spare hours cover delays and bugs. The extras start only after checkpoint 2, so the core is always finished first.

Member	Development	Docs & presentation	Total (of 72 h)
Ibrahim Hassan Anwar (team leader)	≈ 47 h	≈ 12 h	≈ 59 h
Zeyad Mohammed Mohammed	≈ 47 h	≈ 12 h	≈ 59 h
Hager Waleed Shaker	≈ 47 h	≈ 12 h	≈ 59 h
Mohammed Eslam Mohammed	≈ 47 h	≈ 12 h	≈ 59 h
Marawan Hesham Tolba	≈ 47 h	≈ 12 h	≈ 59 h
Khaled Atef Salah	≈ 47 h	≈ 12 h	≈ 59 h

Tools and services (cost: $0).

Need	Tool	Plan
Code and tasks	GitHub, GitHub Projects, GitHub Actions	Free
Frontend hosting	Vercel	Free (Hobby)
Backend hosting	Render (Docker)	Free, kept awake with an uptime ping
Database	MongoDB Atlas	Free (M0)
File storage	Cloudinary	Free tier

Need	Tool	Plan
AI paper check	Mock mode for development, free AI tier for the demo with sample papers only	Free
Design	Figma	Free

At launch, the only real cost would be a paid AI model for real papers (roughly $3 to $12 per 1,000 papers).

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3. Task Assignment & Roles

Roles

All six members have the same amount of work. **Ibrahim Hassan Anwar is the team leader.** Besides his own area, he set up the project and the repository, and he has the final say on scope.

We build the whole **frontend first**, with sample data, and each member owns **one area of the website** from start to finish. Each area has about the same amount of work: one big, one medium and one small page, or the equivalent.

Member	Role	Area	Pages
Ibrahim Hassan Anwar	**Team leader** · Visitors	Public pages	Home, Signup, Login, plus the project setup
Zeyad Mohammed Mohammed	Donor	Donor pages	Nearby requests, Upload certificate, Donation history
Hager Waleed Shaker	Donation	The donation itself	Request details (accept, QR or no QR), Confirm donation, Terms of use + 404
Mohammed Eslam Mohammed	Requester	Requester pages	New request (4 steps + hospital paper), Request progress
Marawan Hesham Tolba	Hospital	Partner hospital pages	Hospital dashboard (QR scan), Confirm papers, Hospital signup
Khaled Atef Salah	Admin	Admin pages	Admin dashboard, Review papers, Reports and bans, Approve hospitals

Each member also builds the shared parts their pages need, and documents and tests their own area. The backend will be split between the members once the frontend is finished.

Documents

The team leader collects each phase document and submits it. Everyone contributes, and each member writes the parts about their own area.

How we work

- **Board:** GitHub Projects, with the columns To do, In progress and Done.
- **Meetings:** a 1-hour planning meeting at the start of each sprint, and a short written update every two days.

- **Code:** one branch per task. Each member checks their work (`npm run lint` and `npm run build`), opens a pull request and squash-merges it into `main`.
- **Decisions:** after sprint 2, only the team leader can add new features to the plan.

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4. Risk Assessment & Mitigation Plan

#	Risk	Likelihood	Impact	Mitigation
1	A team member is unavailable for a week or more	Medium	High	Hours planned for the worst case. Each area is documented and uses the same shared components, so another member can cover it.
2	The work runs out of time	Medium	High	A fixed core scope, checkpoints on 6 and 13 Nov, a feature freeze on 20 Nov and a 3-day buffer.
3	Six people working on the same code cause conflicts or duplicated work	Medium	Medium	Each member owns clear folders and features, pull requests stay small, and there's a short written update every two days.
4	Documentation takes time away from coding	High	Medium	Each phase has one owner, diagrams are written as code in the repo, and documents are written while the feature is being built.
5	The AI marks a fake paper "looks valid"	Medium	Medium	The badge never claims a paper is genuine. Users accept terms saying they are responsible for their papers. Users can report, and abusers are banned. Partner hospitals give the stronger badge.
6	The free AI service is slow or down during the demo	Medium	High	Mock mode (`AI_PROVIDER=mock`), papers checked in advance and a recorded backup video.
7	Push notifications are hard to get working	Medium	Medium	Started early in sprint 3. Live updates on the website are the fallback. Demo on Android Chrome.
8	The free server sleeps and misses notification waves	High	Low	An uptime ping every 5 to 10 minutes.
9	No real hospital agrees to partner before the demo	High	Low	Demo hospitals have clearly fictional names. We never claim a partnership that doesn't exist.
10	Users' personal data or papers leak	Low	High	Papers are private files, phone numbers are hidden until a donor accepts, no secrets go in the repo, and only sample papers are sent to the free AI. We follow Egypt's Personal Data Protection Law (151/2020).
11	People ask for money or misuse the platform	Medium	High	Clear "no money" rules on every screen, a report button and admin bans.
12	Users see us as responsible for medical outcomes	Low	High	Terms of use say we only connect people. The hospital makes every medical decision.

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5. KPIs (Key Performance Indicators)

These are targets we set for the demo and a small pilot. They are not measured results yet.

Product

KPI	Target	How we measure
Time from request to first donor accepting	Under 30 minutes in the demo	Timestamps stored on each request
Requests that reach at least one accepting donor	≥ 70 % in the pilot	Admin dashboard
Notified donors who accept or decline	≥ 50 %	Notification and response logs
Requests that have a hospital paper	100 %	Enforced by the system
Abuse reports handled by an admin	100 % within 24 hours	Report status and timestamps

Technical

KPI	Target	How we measure
API response time	95 % of requests under 500 ms	Server logs
Uptime during demo week	≥ 99 %	Uptime monitor
Page load on a phone over 4G	Under 3 seconds	Lighthouse
Accessibility score	≥ 90	Lighthouse
Test coverage of core logic (matching, trust badges, request states)	≥ 70 %	Jest coverage report

Adoption (pilot)

KPI	Target	How we measure
Registered donors	50	Database count
Donors with an uploaded certificate	≥ 60 % of registered donors	Database count
Partner hospitals	At least 1 real partner talked to	Outreach log

Project delivery

KPI	Target
Phase documents submitted on time	7 of 7
Core features done by checkpoint 2 (13 Nov)	100 %
Pull requests reviewed before merging	100 %
Critical bugs open at submission	0

Stakeholder Analysis

Stakeholder	Who they are	What they need	Influence	How we involve them
Requester	A patient's family member or friend	Find compatible donors fast, without posting everywhere	High	Interviews, usability test of the request flow
Donor	A volunteer who wants to give blood	Only relevant, real requests close to home; privacy	High	Short survey, usability test of the donor flow
Partner hospital	Hospital staff using our portal	Confirm papers quickly, record donations, reach donors for their patients	High	Talk to 1 or 2 hospitals; demo uses fictional ones
Non-partner hospital	Any other official hospital	Nothing from us; it still examines the donor and decides	Low	None. Its papers are checked by AI
Admin	Our moderation team	Review unclear papers, handle reports, ban abusers	Medium	Built by the team
AI service provider	The vision AI API	Clear prompts, sample papers only on the free tier	Low	Through one `checkPaper()` function
DEPI lecturer and evaluators	Grade the project	Clear docs, working demo, good code	High	Phase reviews and the final presentation
Project team	The 6 members	A realistic plan and clear roles	High	Weekly meetings, GitHub board

Database Design

1. Overview

	Choice	Why
Database	MongoDB Atlas (free M0 cluster)	Flexible documents fit papers, AI results and request history. Built-in geo queries for "donors near this hospital". Free for the demo.
Access	Mongoose models in the TypeScript backend (Node + Express)	Schemas, validation and references in one place. Taught in the course.
Files	Cloudinary (private, signed URLs)	Paper images and PDFs are never stored in the database, only their private file reference.
Live updates	Socket.io on top of the database	When a pledge is saved, the requester's page updates without a refresh.

Design rules

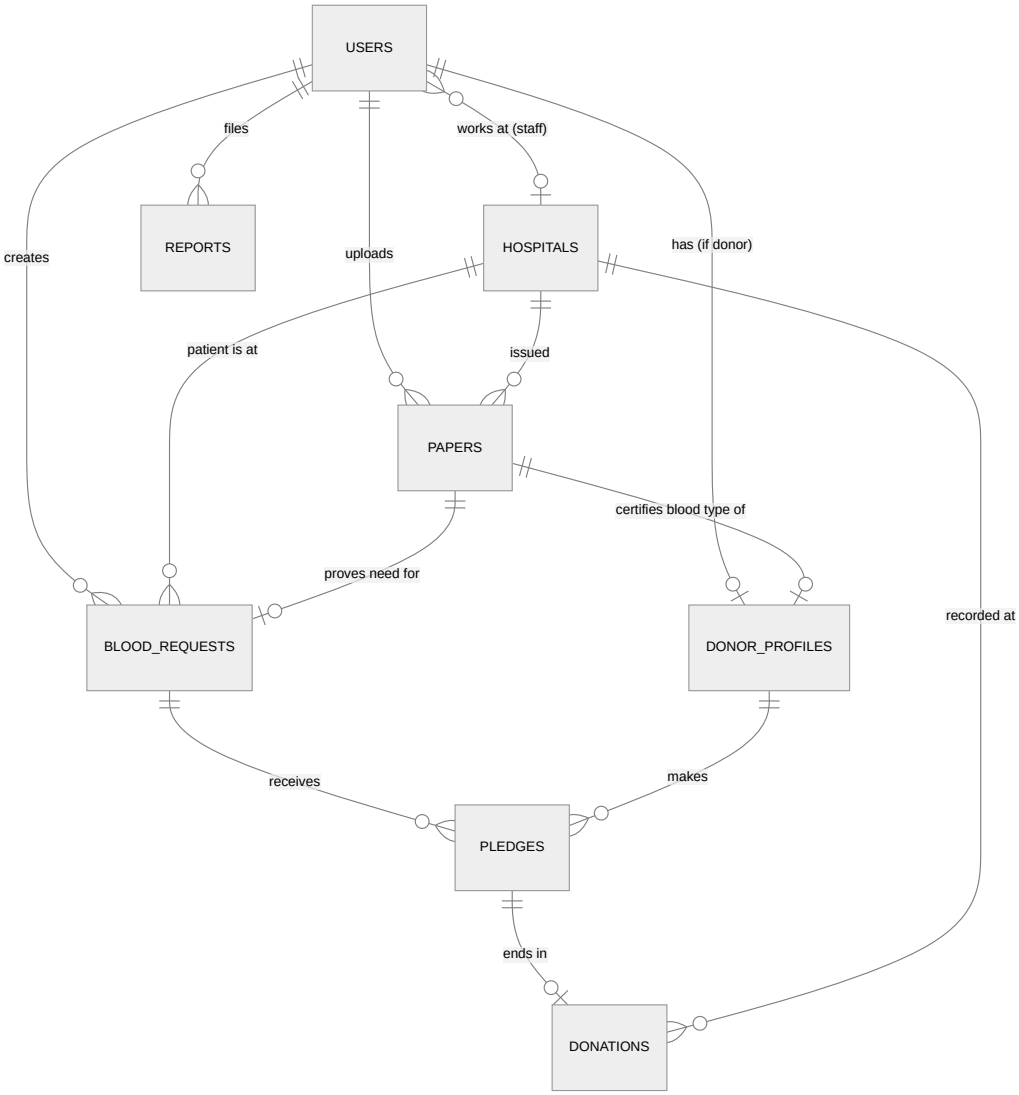
1. **One collection per main thing** a user can point at: a user, a hospital, a paper, a request, a pledge, a donation, a report.
2. **Reference, don't copy,** for things that change (a hospital's name, a user's phone). **Embed** small things that belong to one parent only (a request's history, an AI result).
3. **Every document has a human reference number** (like `BL-2026-004213`) next to its database `_id`, so users and support can talk about the same item.
4. **Private by default.** Phone numbers and papers are only returned by the API in the cases listed in section 8.
5. **Times are stored in UTC.** The website shows them in Cairo time.

Collections at a glance

Collection	Holds
`users`	Every account: name, phone, password hash, roles
`donorProfiles`	Donor-only data: blood type, location, availability, trust
`hospitals`	All hospitals, partner or not
`papers`	Uploaded certificates and hospital papers, with the check result
`bloodRequests`	Requests for blood
`pledges`	A donor's answer to a request (accept / decline), with the QR token
`donations`	What actually happened at the hospital
`notifications`	Who was notified about which request, in which wave
`pushSubscriptions`	Browser push endpoints
`reports`	Reports about users or requests
`auditLogs`	Every admin and hospital-staff action
`settings`	One document: wave radius, pause period, AI mode

Collection	Holds
`counters`	Next number for each reference type

2. Entity relationship diagram



The fields of each collection are listed in section 3. The diagram shows the 8 core collections. The others only point at users and requests: `notifications` (user + request), `pushSubscriptions` (user) and `auditLogs` (the acting user). `settings` and `counters` stand alone.

Reading the main relationships

- A **user** can be a donor, a requester, or both. Only donors have a **donor profile** (one each).
- Hospital **staff** are users with the `hospital_staff` role and a `hospitalId`.
- A **request** belongs to one requester and one hospital, and is backed by exactly one **paper**.
- A **pledge** links one donor to one request. A request has many pledges, one per donor who answered.
- A pledge ends in at most one **donation** record: donated, not today, didn't show up, or self-confirmed.

3. Collections

Every collection also has `createdAt` and `updatedAt` (added automatically by Mongoose).

3.1 `users`

Field	Type	Rules	Notes
`_id`	ObjectId		
`name`	string	required, 3–80 chars	Full name in Arabic
`phone`	string	required, unique, Egyptian mobile `01XXXXXXXXX`	**Private** (section 8)
`email`	string	optional, unique when present, lowercase	Login with email or phone
`passwordHash`	string	required	bcrypt, never returned by the API
`roles`	string[]	at least 1 of `donor`, `requester`, `hospital_staff`, `admin`	Donor + requester together is allowed
`hospitalId`	ObjectId → `hospitals`	required if `hospital_staff`	
`termsAcceptedAt`	date	required	When the user accepted the terms of use
`aiConsentAt`	date	optional	Consent before a paper goes to the AI check
`status`	string	`active`, `warned`, `banned`	Banned users can't log in
`ban`	object	`{ reason, by, at }`	Set by an admin
`lastLoginAt`	date		

3.2 `donorProfiles`

Field	Type	Rules	Notes
`userId`	ObjectId → `users`	required, unique	One profile per donor
`ref`	string	unique, `DN-2026-01877`	Shown on the donor's ID card
`bloodType`	string	one of the 8 blood types	From the certificate
`governorate`, `area`	string	required	For display and filters
`location`	GeoJSON Point	required, `[longitude, latitude]`	Used for distance matching
`isAvailable`	boolean	default `true`	The availability switch
`pausedUntil`	date	optional	Set after a donation or by the donor
`trustLevel`	string	`pending`, `needs_review`, `looks_valid`, `verified`	Copied from the certificate's result
`certificatePaperId`	ObjectId → `papers`		The current certificate

Field	Type	Rules	Notes
`stats`	object	`{ donations, noShows, lastDonationAt }`	Kept up to date when a donation is recorded

3.3 `hospitals`

Field	Type	Rules	Notes
`ref`	string	unique, `HP-0007`	
`name`	string	required	e.g. "مستشفى النور"
`type`	string	`public`, `private`, `university`, `military`	
`governorate`, `area`, `address`	string	required	
`location`	GeoJSON Point	required	Matching measures distance from here
`isPartner`	boolean	default `false`	Only partners can confirm papers and scan QR codes
`partnerStatus`	string	`none`, `pending`, `approved`, `rejected`	Partner accounts wait for admin approval
`partnerRequest`	object	`{ contactName, contactPhone, documentPaperId, submittedAt }`	From the hospital signup form
`approvedBy`, `approvedAt`	ObjectId → `users`, date		
`bloodBank`	object	`{ building, floor, hours, phone }`	Directions shown to donors after accepting

Non-partner hospitals are also stored, so requests can point at them and donors see the address.

3.4 `papers`

One document per uploaded file: a donor's blood-type certificate or a requester's hospital paper.

Field	Type	Rules	Notes
`ownerId`	ObjectId → `users`	required	Who uploaded it
`kind`	string	`donor_certificate`, `need_paper`, `hospital_license`	
`hospitalId`	ObjectId → `hospitals`	required	The hospital that issued it
`fromPartner`	boolean		Copied from the hospital at upload time
`file`	object	`{ publicId, mime, sizeBytes }`; JPG, PNG or PDF, ≤ 5 MB	Cloudinary private file. **Never public**
`ai`	object	`{ provider, result, extracted, problems, checkedAt }`	Only for non-partner papers

Field	Type	Rules	Notes
`ai.extracted`	object	`{ name, bloodType, stampVisible, issueDate }`	What the AI could read
`status`	string	see 5.2	
`review`	object	`{ by, at, decision, note }`	Partner staff or admin decision
`expiresAt`	date		Need papers are only valid for a short time

3.5 `bloodRequests`

Field	Type	Rules	Notes
`ref`	string	unique, `BL-2026-004213`	Used in the status lookup and WhatsApp share
`createdBy`	ObjectId → `users`	required	The requester
`patient`	object	`{ name, age, department }`	Shown only to accepted donors and hospital staff
`bloodType`	string	one of the 8 blood types	The patient's type
`component`	string	`whole_blood`, `platelets`, `plasma`	
`unitsNeeded`	number	1–10	
`unitsPledged`	number	≥ 0	Accepted pledges still open. Kept in sync for the live counter
`unitsDonated`	number	≥ 0	
`urgency`	string	`urgent`, `normal`	
`neededBy`	date	required, in the future	The request expires after this
`hospitalId`	ObjectId → `hospitals`	required	
`paperId`	ObjectId → `papers`	required	Can't be published without it
`trustLevel`	string	copied from the paper	The badge donors see
`status`	string	see 5.1	
`wave`	object	`{ number, radiusKm, startedAt, nextAt }`	1 = 5 km, 2 = 15 km
`history`	array of `{ at, event, by }`	embedded, append-only	The timeline on the progress page

3.6 `pledges`

Created when a donor answers a request.

Field	Type	Rules	Notes
`ref`	string	unique, `AP-2026-118204`	Shown on the donor's ticket

Field	Type	Rules	Notes
`requestId`	ObjectId → `bloodRequests`	required	
`donorId`	ObjectId → `donorProfiles`	required	One pledge per donor per request (unique pair)
`status`	string	`accepted`, `declined`, `cancelled`, `closed`	
`wave`	number		Which wave reached this donor
`distanceKm`	number		Donor to hospital, at the time of matching
`qrToken`	string	unique, random, only for partner hospitals	Encoded in the QR code
`qrExpiresAt`	date		Valid until the request's deadline
`acceptedAt`, `answeredAt`	date		`acceptedAt` unlocks phone numbers

3.7 `donations`

Field	Type	Rules	Notes
`pledgeId`	ObjectId → `pledges`	required, unique	
`requestId`, `donorId`, `hospitalId`	ObjectId	required	Copied for fast history and statistics queries
`outcome`	string	`donated`, `deferred`, `no_show`, `self_confirmed`	`deferred` = "not able to donate today (medical decision)"
`method`	string	`qr_scan`, `self_confirmation`	
`recordedBy`	ObjectId → `users`	for `qr_scan`	The hospital staff member
`confirmations`	object	`{ donorAt, requesterAt }`	For `self_confirmation`: both must confirm
`staffNote`	string	optional, ≤ 200 chars	Internal, admin only. **No medical details.**
`at`	date	required	

3.8 `notifications`

Field	Type	Notes
`userId`	ObjectId → `users`	The donor
`requestId`	ObjectId → `bloodRequests`	
`wave`	number	So the same donor is never notified twice for one request
`channel`	string	`push`, `in_app`
`sentAt`, `readAt`	date	

3.9 `pushSubscriptions`

Field	Type	Notes
`userId`	ObjectId → `users`	One user can have several browsers
`endpoint`	string, unique	From the browser's Push API
`keys`	object `{ p256dh, auth }`	
`failedAt`	date	Removed after repeated failures

3.10 `reports`

Field	Type	Rules	Notes
`ref`	string	unique, `RP-2026-000412`	
`reporterId`	ObjectId → `users`	required	
`target`	object	`{ kind: 'user' | 'request', id }`	Report button on every profile and request
`reason`	string	`money_request`, `fake_paper`, `harassment`, `no_show`, `other`	Asking for money = highest priority
`details`	string	≤ 500 chars	
`status`	string	`open`, `dismissed`, `warned`, `banned`	What the admin decided
`handledBy`, `handledAt`	ObjectId, date		

3.11 `auditLogs`

Every admin and hospital-staff action is logged with the actor's name: paper decisions, bans, hospital approvals, QR scans.

Field	Type	Notes
`actorId`	ObjectId → `users`	
`action`	string	e.g. `paper.approve`, `user.ban`, `donation.record`
`target`	object `{ collection, id, ref }`	
`at`	date	
`details`	object	Short, no personal data beyond the reference

3.12 `settings` and `counters`

`settings` holds **one document** that admins can change without a new deploy:

Field	Default	Used for
`waveRadiiKm`	`[5, 15]`	Wave 1 and wave 2 search radius
`waveDelayMinutes`	`120`	Time before the next wave starts
`pauseDaysAfterDonation`	`90`	General guidance; the hospital decides

Field	Default	Used for
`aiProvider`	`mock`	`mock` or a real provider
`needPaperMaxAgeDays`	`7`	How recent a hospital paper must be

`counters` makes reference numbers: `{ _id: 'BL-2026', seq: 4213 }`. The backend increments it atomically with `findOneAndUpdate({ $inc: { seq: 1 } })`.

4. Shared values (enums)

These lists are shared by the database, the API and the frontend constants, so a value is spelled the same everywhere.

Name	Values
Blood types	`O-`, `O+`, `A-`, `A+`, `B-`, `B+`, `AB-`, `AB+`
Roles	`donor`, `requester`, `hospital_staff`, `admin`
Trust levels	`verified`, `looks_valid`, `needs_review`, `pending`
Request status	`draft`, `checking`, `pending_hospital`, `needs_review`, `active`, `fulfilled`, `expired`, `rejected`, `cancelled`
Pledge status	`accepted`, `declined`, `cancelled`, `closed`
Donation outcome	`donated`, `deferred`, `no_show`, `self_confirmed`
Report status	`open`, `dismissed`, `warned`, `banned`

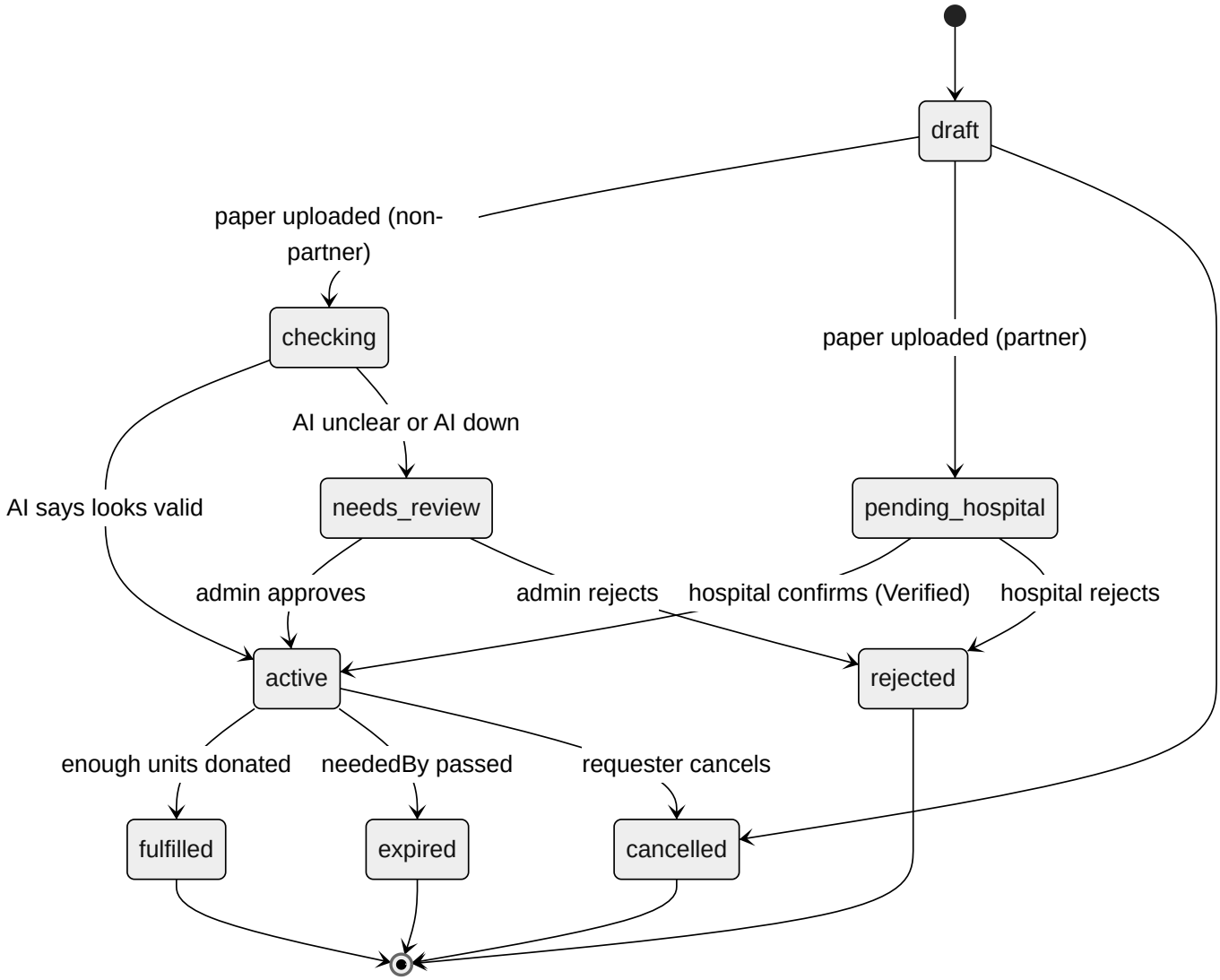
Blood compatibility (used by matching). Which donors can give to a patient:

Patient	Compatible donors
O−	O−
O+	O−, O+
A−	O−, A−
A+	O−, O+, A−, A+
B−	O−, B−
B+	O−, O+, B−, B+
AB−	O−, A−, B−, AB−
AB+	All 8 types

This table lives in code (a constant), not in the database, because it never changes.

5. Status lifecycles

5.1 Blood request



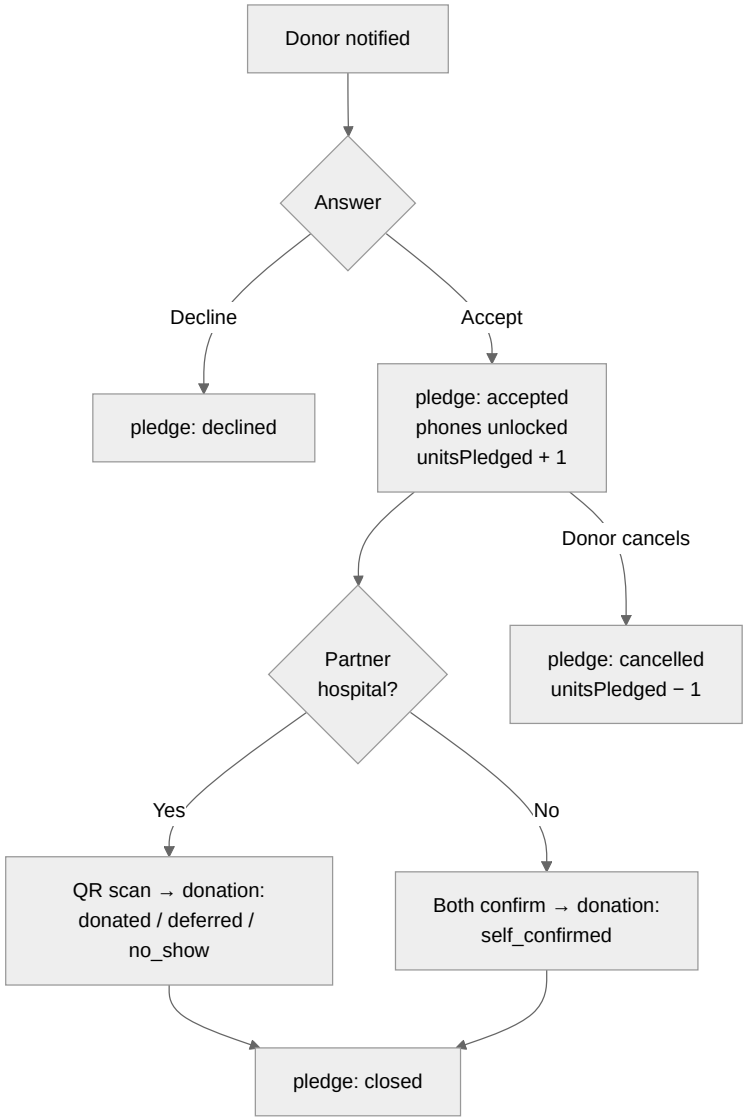
Only `active` requests are matched and shown to donors.

5.2 Paper and trust level

Paper status	When	Trust level given
`pending_hospital`	Uploaded, issued by a partner hospital	`pending`
`verified`	The partner hospital confirmed "issued by us"	`verified` (**Verified**)
`looks_valid`	Non-partner, the AI read it and found no problems	`looks_valid` (**Paper looks valid**)
`needs_review`	The AI couldn't read it, or the AI service is down	`needs_review`
`rejected`	Hospital or admin rejected it	(request can't be published)

A self-confirmed donation at a non-partner hospital **never** upgrades a donor to `verified`. Only a QR scan at a partner hospital can.

5.3 Pledge and donation



6. Indexes

Collection	Index	Why
`users`	`{ phone: 1 }` unique	Login and duplicate check
`users`	`{ email: 1 }` unique, sparse	Email is optional
`donorProfiles`	`{ location: '2dsphere' }`	Find donors within 5 / 15 km
`donorProfiles`	`{ bloodType: 1, isAvailable: 1, pausedUntil: 1 }`	Filter matching donors fast
`donorProfiles`	`{ userId: 1 }` unique, `{ ref: 1 }` unique	
`hospitals`	`{ location: '2dsphere' }`, `{ ref: 1 }` unique, `{ isPartner: 1 }`	Nearby requests, partner list
`bloodRequests`	`{ ref: 1 }` unique	Status lookup by reference
`bloodRequests`	`{ status: 1, neededBy: 1 }`	Active list and the expiry job
`bloodRequests`	`{ createdBy: 1, createdAt: -1 }`	"My requests"

Collection	Index	Why
`bloodRequests`	`{ hospitalId: 1, status: 1 }`	Hospital dashboard
`papers`	`{ status: 1, kind: 1, createdAt: 1 }`	Admin and hospital queues, oldest first
`papers`	`{ hospitalId: 1, status: 1 }`	"Papers waiting" for one hospital
`pledges`	`{ requestId: 1, donorId: 1 }` unique	One answer per donor per request
`pledges`	`{ donorId: 1, status: 1 }`	Donor's appointments
`pledges`	`{ qrToken: 1 }` unique, sparse	QR scan lookup
`donations`	`{ donorId: 1, at: -1 }`	Donation history
`donations`	`{ hospitalId: 1, at: -1 }`	Hospital's log
`notifications`	`{ requestId: 1, userId: 1 }` unique	Never notify twice
`reports`	`{ status: 1, reason: 1, createdAt: 1 }`	Admin queue, money requests first
`auditLogs`	`{ at: -1 }`, `{ actorId: 1, at: -1 }`	

7. Key queries

Matching: donors for a request. One geo query, run per wave:

```
db.donorProfiles.aggregate([
  { $geoNear: {
      near: hospital.location,               // the request's hospital
      distanceField: 'distanceM',
      maxDistance: wave.radiusKm * 1000,     // 5 km, then 15 km
      query: {
        bloodType: { $in: COMPATIBLE_DONORS[request.bloodType] },
        isAvailable: true,
        $or: [{ pausedUntil: null }, { pausedUntil: { $lt: now } }],
        trustLevel: { $in: ['verified', 'looks_valid'] },
      },
  } },
  // then: remove banned users, donors already notified, and the requester themself
  { $limit: 50 },
]);
```

Closest and most trusted donors come first. Banned users and donors already in `notifications` for this request are filtered out.

Live counter. When a pledge is accepted or cancelled, `unitsPledged` is changed with `$inc` in the same operation, and Socket.io sends the new number to the request's room.

Expiry job. Every 10 minutes: `updateMany({ status: 'active', neededBy: { $lt: now } }, { $set: { status: 'expired' } })`.

Admin key numbers. Counts on `bloodRequests` (active), `donorProfiles`, `reports` (open), and the average time from a request's `createdAt` to its first pledge's `acceptedAt`.

8. Privacy and security rules

The database stores personal data, so the API decides exactly who can see what (Egypt's Personal Data Protection Law 151/2020).

Data	Who can see it
Donor's phone	The requester, **only after** that donor accepted their request
Requester's phone	A donor, **only after** they accepted that request
Paper files	The owner, the issuing partner hospital's staff (to confirm it), and admins. **Never** other users
Patient name and department	Accepted donors of that request, and hospital staff
`passwordHash`, `qrToken`	Never returned, except the donor's own QR token on their ticket
`staffNote` on donations	Admins only
Reports	The reporter and admins. The reported user never sees who reported them

Other rules

- Passwords are hashed with bcrypt. Login uses a JWT in an httpOnly cookie.
- Every input is validated with zod before it reaches Mongoose.
- Connection strings and keys live only in `.env`, never in the repo.
- Only sample papers are sent to the free AI tier.
- Database users: the app connects with a user that can only read and write the `bloodlink` database.

9. Validation and data rules

Rule	Where
Phone matches `^01[0125][0-9]{8}$` (Egyptian mobile)	zod + Mongoose
Files: JPG, PNG or PDF, max 5 MB	Upload middleware
A request needs `paperId`, and `neededBy` in the future	zod + Mongoose
`unitsPledged` ≤ `unitsNeeded` + 1 (one spare donor allowed)	Service layer, in the same update
One pledge per donor per request	Unique index
A donor can't pledge to their own request	Service layer
A donor who is paused, unavailable or banned is never matched	Matching query
Deleting is soft: banned users are kept (with `status: 'banned'`), so reports and history stay correct	Service layer
Reference numbers are never reused	`counters` collection

Retention. Paper files are deleted from Cloudinary 90 days after the request ends. Request and donation records are kept for history and statistics, without the files.

10. Example documents

A request (`bloodRequests`):

```
{
  "_id": "6721f3a0c2b1a4e5f0a10001",
  "ref": "BL-2026-004213",
  "createdBy": "6721f2ffc2b1a4e5f0a0aaa1",
  "patient": { "name": "سارة إبراهيم", "age": 54, "department": "قسم الجراحة" },
  "bloodType": "O+",
  "component": "whole_blood",
  "unitsNeeded": 3,
  "unitsPledged": 2,
  "unitsDonated": 0,
  "urgency": "urgent",
  "neededBy": "2026-10-23T03:00:00Z",
  "hospitalId": "6721f100c2b1a4e5f0a00007",
  "paperId": "6721f390c2b1a4e5f0a0f001",
  "trustLevel": "verified",
  "status": "active",
  "wave": { "number": 1, "radiusKm": 5, "startedAt": "2026-10-22T07:12:00Z" },
  "history": [
    { "at": "2026-10-22T07:05:00Z", "event": "created" },
    { "at": "2026-10-22T07:10:00Z", "event": "paper_verified", "by": "6721f2aac2b1a4e5f0a0bbb2"
},
    { "at": "2026-10-22T07:12:00Z", "event": "wave_started", "wave": 1 }
  ]
}
```

A pledge (`pledges`) at a partner hospital:

```
{
  "_id": "6721f5d0c2b1a4e5f0a20001",
  "ref": "AP-2026-118204",
  "requestId": "6721f3a0c2b1a4e5f0a10001",
  "donorId": "6721f222c2b1a4e5f0a0d001",
  "status": "accepted",
  "wave": 1,
  "distanceKm": 2.4,
  "qrToken": "q_8f3a1c9e7b2d4f60",
  "qrExpiresAt": "2026-10-23T03:00:00Z",
  "acceptedAt": "2026-10-22T07:52:00Z"
}
```

A donor profile (`donorProfiles`):

```
{
  "_id": "6721f222c2b1a4e5f0a0d001",
  "userId": "6721f1eec2b1a4e5f0a0ccc1",
  "ref": "DN-2026-01877",
  "bloodType": "O+",
  "governorate": "القاهرة",
  "area": "مدينة نصر",
  "location": { "type": "Point", "coordinates": [31.3301, 30.0561] },
  "isAvailable": true,
  "pausedUntil": null,
  "trustLevel": "verified",
  "certificatePaperId": "6721f230c2b1a4e5f0a0f0c1",
  "stats": { "donations": 4, "noShows": 0, "lastDonationAt": "2026-06-07T09:20:00Z" }
}
```


All examples use fictional people and hospitals.

UI/UX Design

1. Design goals

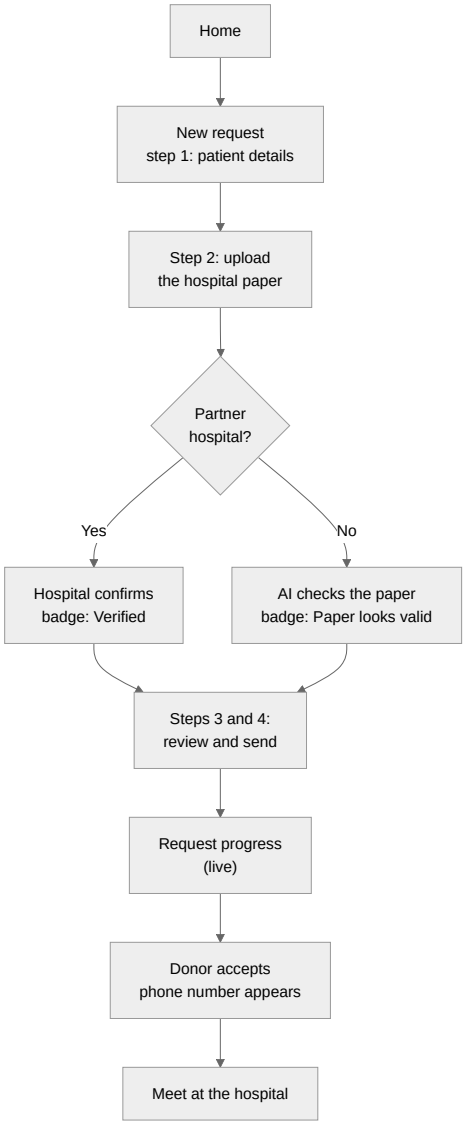
People open BloodLink at a stressful moment: a relative needs blood, often today. The design has to feel **official enough to trust** and **simple enough to use in a hurry**, on a cheap phone, in Arabic.

Goal	What it means in the design
Trust first	Every request and donor shows a trust badge (**Verified**, **Paper looks valid**, **Needs review**). A calm navy-and-gold "public service" look, a seal logo, and reference numbers on every request.
Fast under stress	One main action per screen. Big buttons. The request form is split into 4 short steps, and the draft saves itself.
Safety built in	Phone numbers stay hidden until the donor accepts. "Meet only inside the hospital" and "No money, ever" appear on every key screen.
We connect, the hospital decides	No medical wording or decisions in the UI. Outcomes like "Not able to donate today" are labelled as the hospital's medical decision.
Arabic first	Right-to-left layout, Arabic fonts made for screens, numbers and codes kept left-to-right where needed.
For everyone	Text size, high contrast and dark mode on every page. Works with a keyboard and a screen reader. Works from 360 px wide.

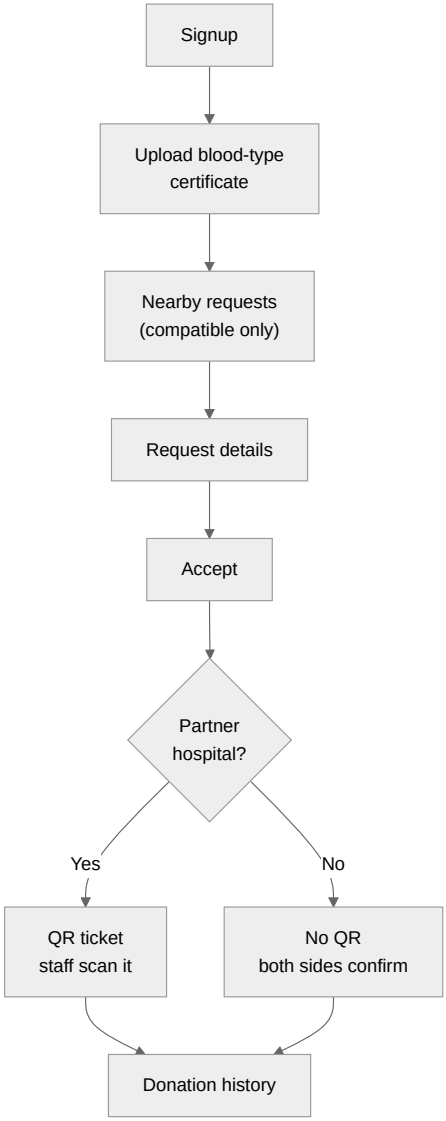
2. Users and key flows

User	Main goal	Main screens
Requester (a patient's family member)	Find compatible donors fast	New request (4 steps), Request progress, Confirm donation
Donor	See real requests near home and help	Nearby requests, Request details, Certificate upload, Donation history
Partner hospital staff	Confirm papers and record donations	Hospital dashboard (QR scan), Papers waiting for confirmation
Admin	Keep the platform safe	Admin dashboard, Unclear papers, Reports and bans, Hospital approvals

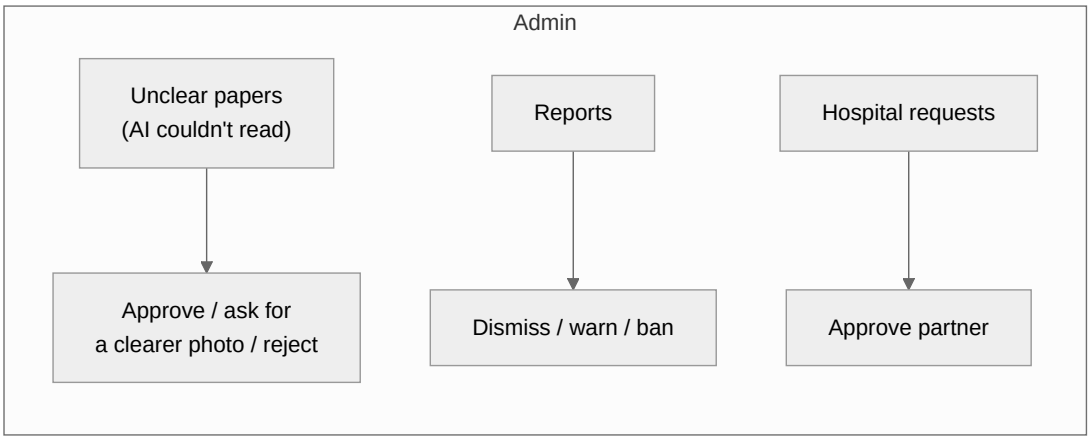
Requester flow

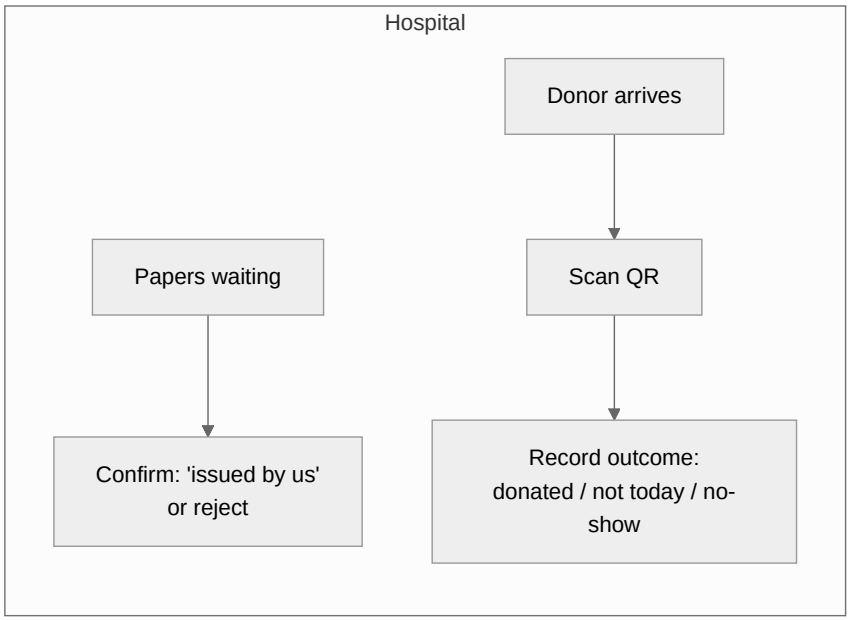


Donor flow


Signup
Upload blood-type certificate
Nearby requests (compatible only)
Request details
Accept
Partner hospital?
Yes
No
QR ticket staff scan it
No QR both sides confirm
Donation history

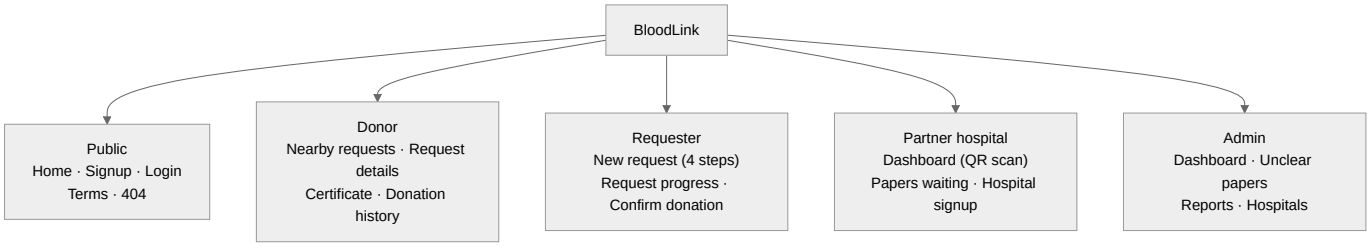
Hospital and admin flows





3. Information architecture

The site has **18 screens** in 5 areas. Each user sees only their own area after login.



Area	Screen	Route
Public	Home	`/`
Public	Signup	`/signup`
Public	Login	`/login`

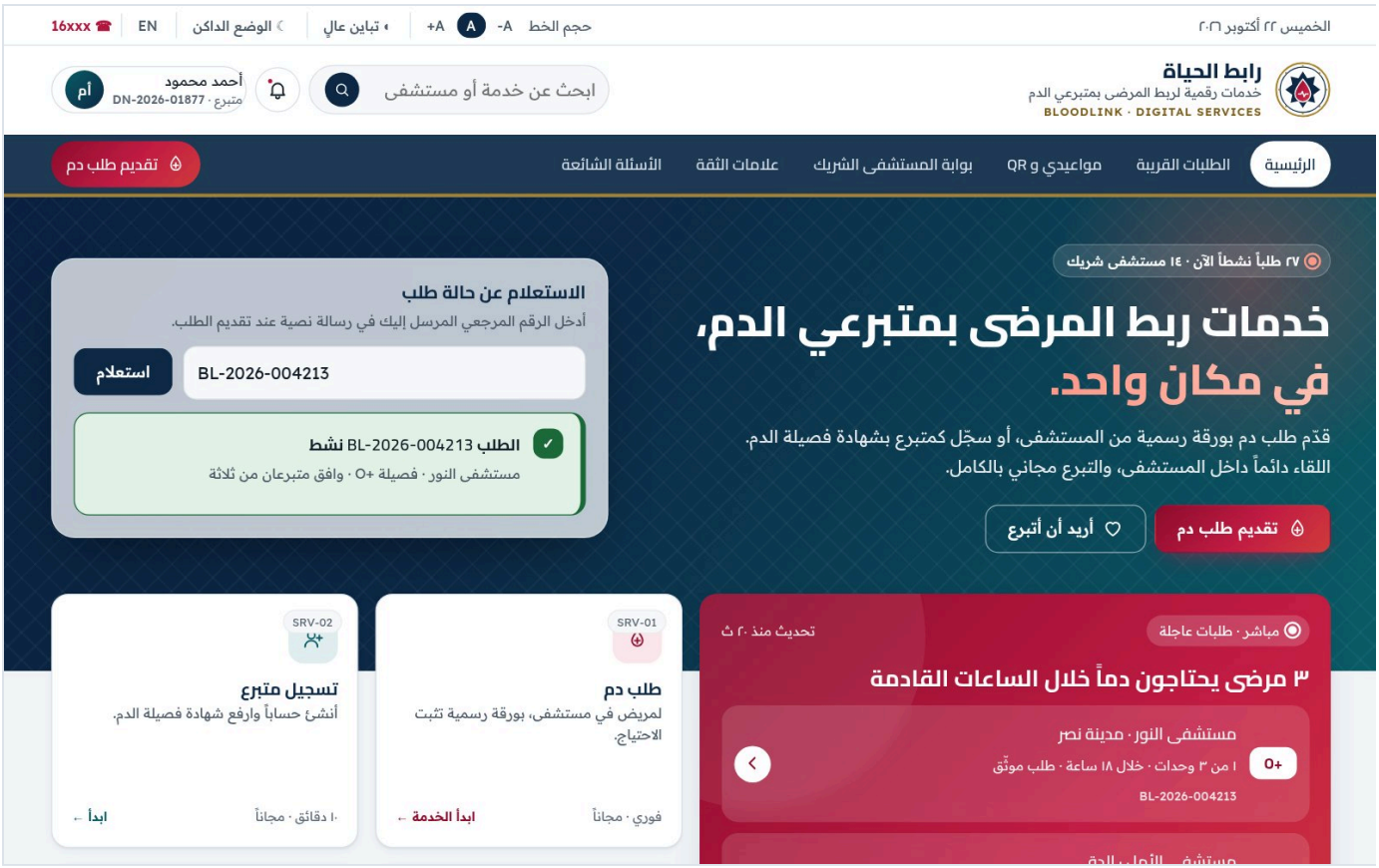
Area	Screen	Route
Public	Terms of use + 404	`/terms`, `*`
Donor	Nearby requests + availability switch	`/donor/requests`
Donor	Request details → accept → QR or no QR	`/donor/requests/:id`
Donor	Blood-type certificate	`/donor/certificate`
Donor	Donation history	`/donor/history`
Requester	New request (4 steps + paper check)	`/requests/new/:step`
Requester	Request progress (live)	`/requests/:id`
Requester	Confirm donation, expired request, WhatsApp share	`/requests/:id/confirm`
Hospital	Dashboard: scan QR, record outcome	`/hospital`
Hospital	Papers waiting for confirmation	`/hospital/papers`
Hospital	Partner signup + waiting for approval	`/hospital/join`
Admin	Dashboard (key numbers)	`/admin`
Admin	Unclear papers	`/admin/papers`
Admin	Reports and bans	`/admin/reports`
Admin	Approve partner hospitals	`/admin/hospitals`

Navigation. On a computer: a utility bar (text size, contrast, dark mode, language), the header with the seal and search, and a main menu with a red **Request blood** button. On a phone: the same header, and a bottom tab bar with 5 tabs (Home, Requests, Request blood, My appointment, Hospital). Inside each area, a side menu lists that area's pages.

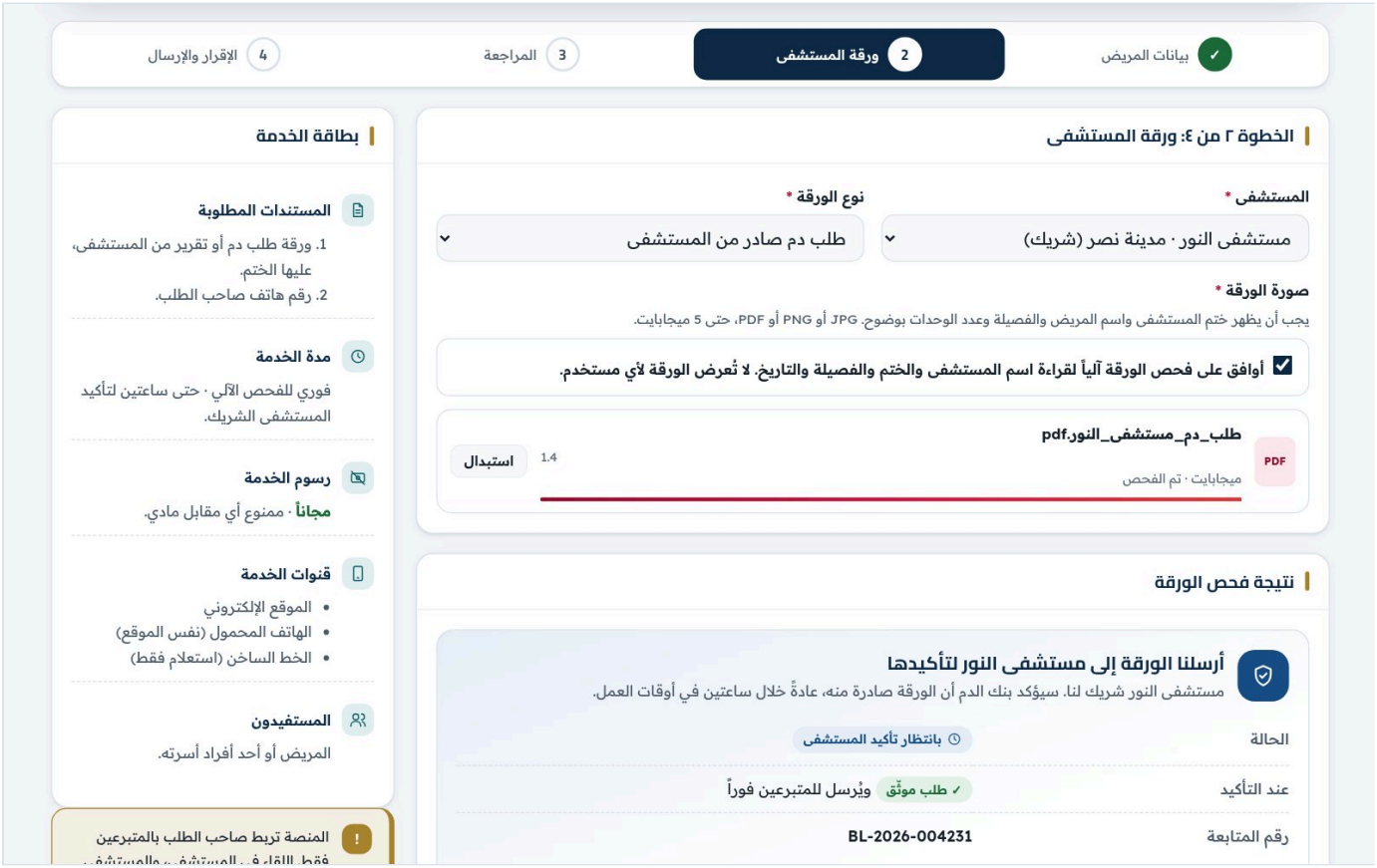
4. Screens

The screens below show the planned design, in Arabic, at real size.

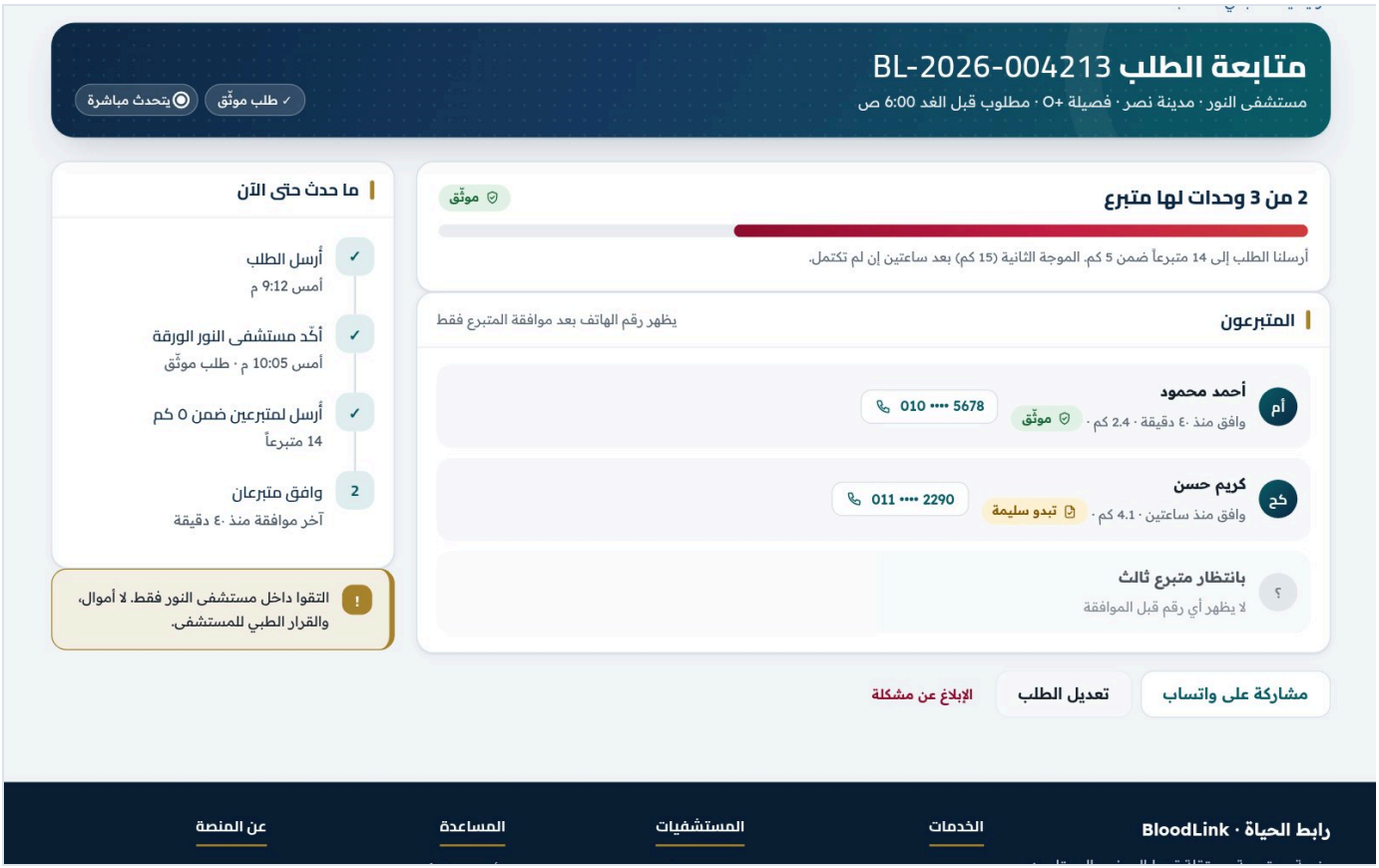
Home (desktop). A short promise, the two main actions (request blood, become a donor), a status lookup by reference number, live urgent requests, and the trust levels explained.



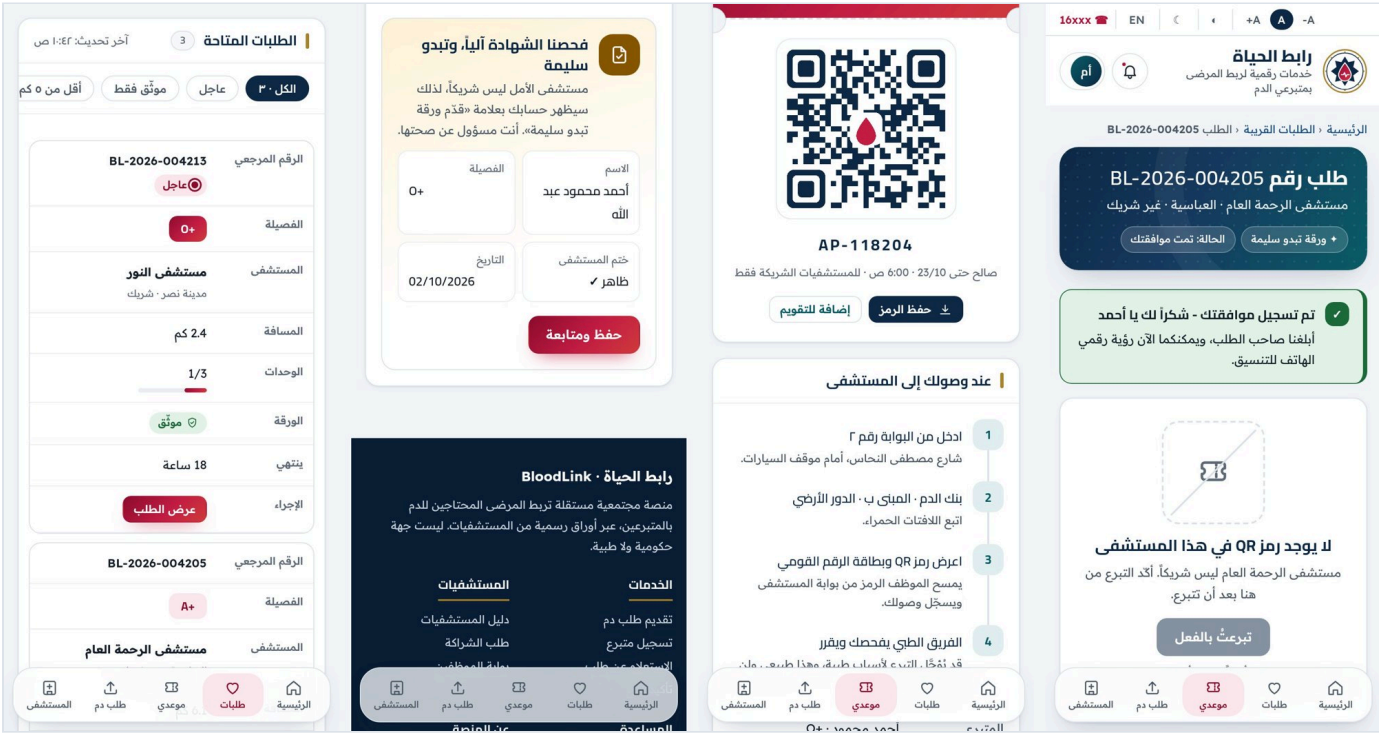
New request, step 2: the hospital paper (desktop). A 4-step form. The paper is sent to the partner hospital, and the result panel says what happens next.



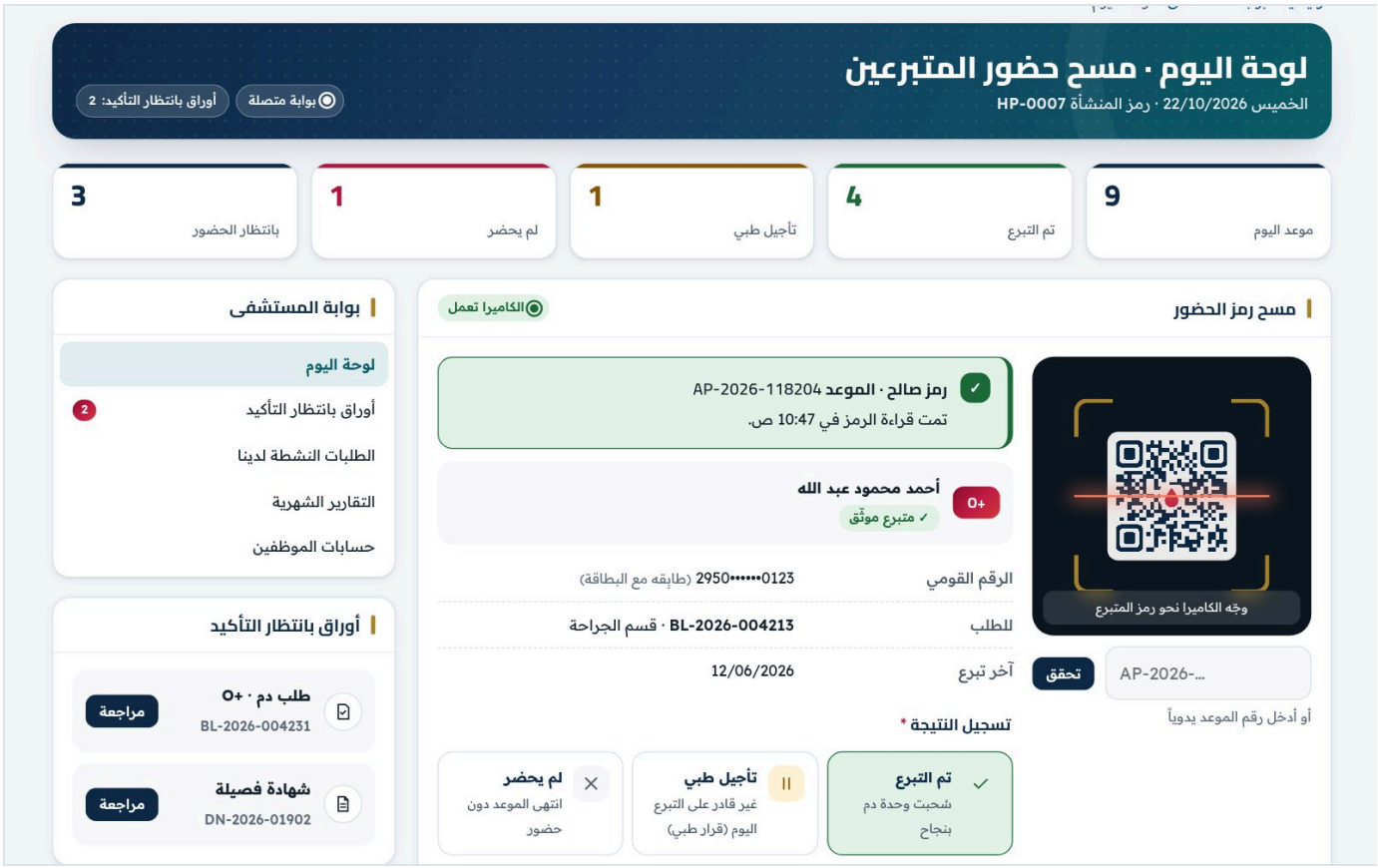
Request progress (desktop). "2 of 3 units have a donor", a timeline of what happened, and donors' phone numbers, shown only after they accept.



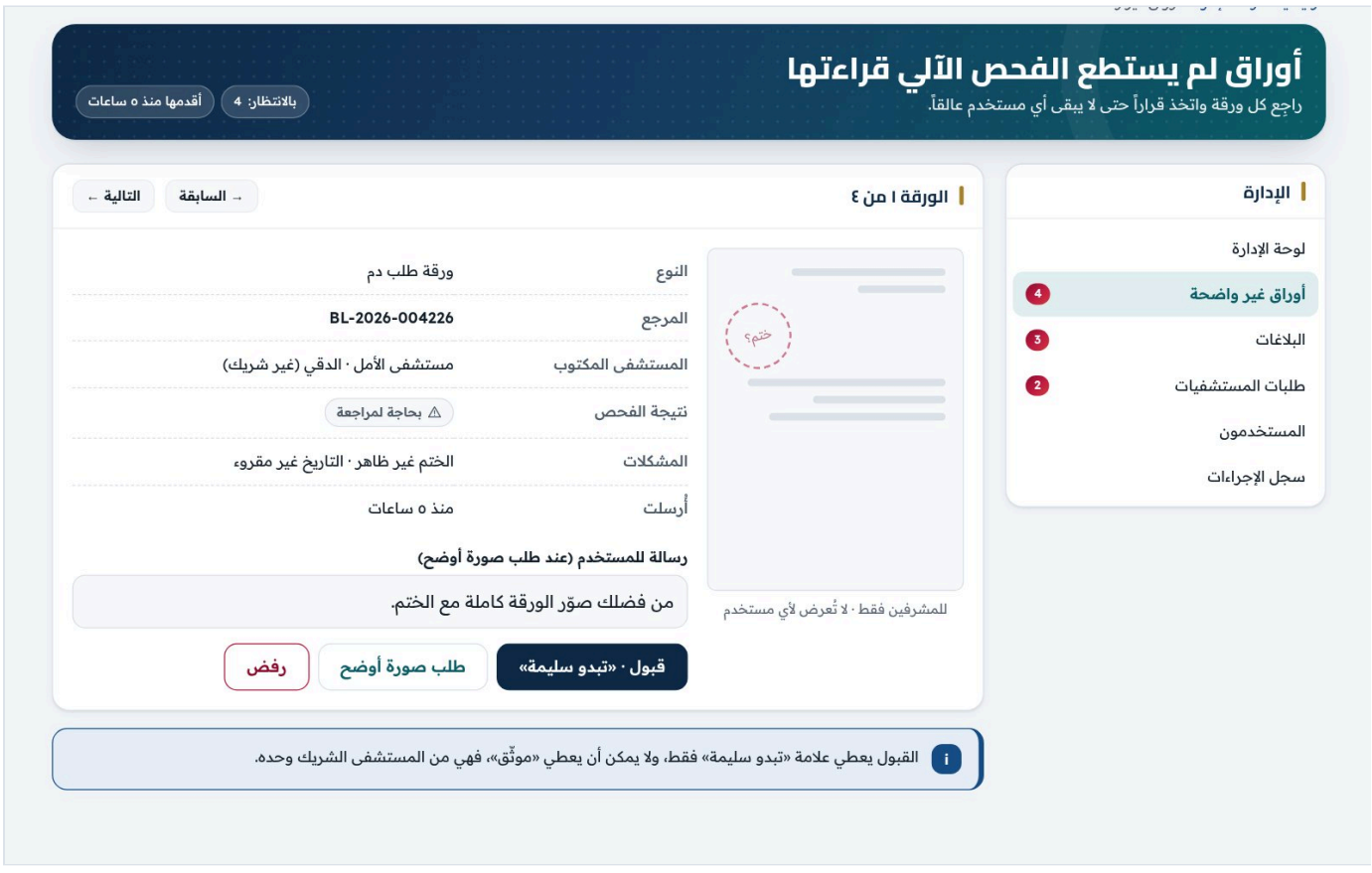
Donor journey (phone). Nearby requests → certificate checked by AI ("looks valid") → QR ticket at a partner hospital → "No QR at this hospital" at other hospitals.



Partner hospital dashboard (desktop). Today's numbers, a QR scanner, and three big outcome buttons: Donated, Not able to donate today (medical decision), Didn't show up.



Admin: unclear papers (desktop). One paper at a time, with what the AI could read and the problems it found, and three actions.



5. Visual design system

The style is a **modern public-service look**: navy and gold for trust (like a government service), crimson only for blood and urgency, lots of white space, rounded cards.

Colours

Every colour is a named **token**. Components never use a raw hex value, so dark mode and high contrast work everywhere automatically. All text colours pass WCAG AA contrast.

Token	Light	Dark	High contrast	Used for
`canvas`	`#F2F5F8`	`#0A131D`	`#FFFFFF`	Page background
`surface`	`#FFFFFF`	`#111E2B`	`#FFFFFF`	Cards, panels, inputs
`ink`	`#15202B`	`#E6EDF4`	`#000000`	Body text
`ink-2`	`#4D5A68`	`#A9B6C4`	`#000000`	Secondary text
`navy`	`#0E2A47`	`#173B60`	`#000000`	Structure: menu, table heads, headings
`teal`	`#0B5E68`	`#127A86`	`#000000`	Accents, selected menu item
`gold`	`#A9842F`	`#A9842F`	`#A9842F`	Decoration only: seal, rules, scanner corners
`crimson`	`#B3163E`	`#FF5A77`	`#8B0000`	Urgent, errors, blood types, main action
`ok`	`#1D6B3A`	`#4FD08F`	`#004D1A`	**Verified**, success
`warn`	`#865600`	`#FFB54D`	`#5A3A00`	**Paper looks valid**, warnings
`info`	`#174E86`	`#8CB4FF`	`#00337A`	Information, links, pending
`focus`	`#F2B705`	`#F2B705`	`#F2B705`	Keyboard focus ring

The main buttons and hero use one gradient: `#8E0D2F → #C41E45 → #D13D3D`.

Colour is never the only signal. Trust badges always have an icon and a word, errors always have text, and blood types are always written out.

Typography

Two Google fonts made for Arabic screens: **Cairo** for headings and **Readex Pro** for text.

Style	Font	Size	Example
Hero title	Cairo 800	1.85 → 2.5 rem	خدمات ربط المرضى بمتبرعي الدم
Page title	Cairo 800	1.45 → 1.8 rem	طلبات قريبة تناسب فصيلتك
Section title	Cairo 800	1.15 rem	المستشفيات الشريكة
Card title	Cairo 700	1–1.08 rem	نتيجة فحص الورقة
Body	Readex Pro 400	1 rem (16 px), line height 1.75	اللقاء دائماً داخل المستشفى
Small	Readex Pro 400–600	0.88 rem	لن نشارك رقم هاتفك إلا بعد موافقتك
Reference codes	Readex Pro 600, left-to-right	inherit	BL-2026-004213

Body text is never smaller than 16 px, and the line height is generous because Arabic letters need room above and below.

Shape and spacing

Element	Radius	Used for
Field	12 px	Inputs, buttons
Tile	14 px	Notices, stats, small cards
Card	16 px	Cards and panels
Hero	22 px	Page header, lookup box

Spacing follows a 4 px scale (4, 8, 12, 16, 24, 32, 40). Two soft shadows: one for cards and one for raised elements.

Icons: one outline icon set (Lucide), always with a text label next to it, and hidden from screen readers when the label already says it.

6. Components

The UI is built from a small set of shared components, so every page looks and behaves the same.

Group	Components
Layout	UtilityBar (text size, contrast, dark mode), Header, MainNav, BottomTabBar (phones), Breadcrumb, PageHeader, SideMenu, Footer
Basics	Button (primary, navy, line, ghost, link), TrustBadge, BloodTypeChip, Notice (info, success, warning, error), Toast
Forms	TextField, SelectField, Checkbox, Switch, FileUpload (with progress), Stepper
Data	DataTable (becomes cards on phones), Pager, KeyValueTable, Skeleton, EmptyState, ErrorState, PledgeProgress
Dialogs	BottomSheet, ConfirmDialog, ReportDialog

Trust badges are the most important component. They look the same everywhere:

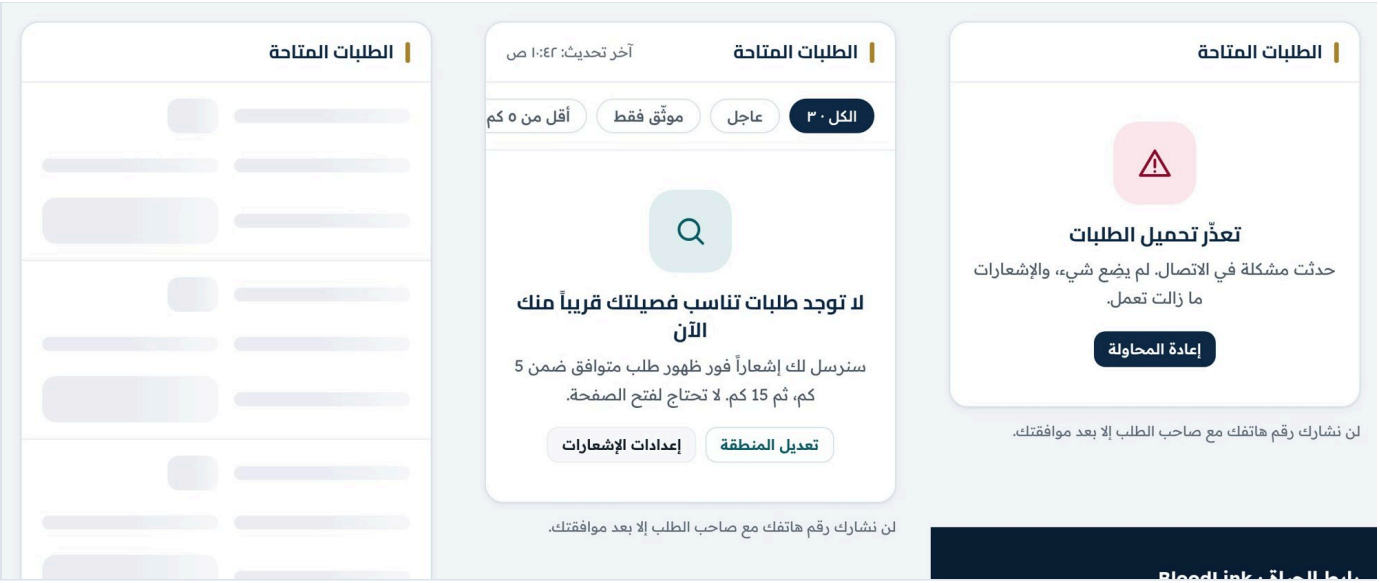
Badge	Colour	Meaning
✓ **Verified** (موثّق)	Green	The paper was issued by a partner hospital, and the hospital confirmed it
Paper looks valid (ورقة تبدو سليمة)	Amber	From another official hospital; the AI checked it. Not a guarantee
Needs review (بحاجة لمراجعة)	Grey	The AI couldn't read it; an admin will check it

7. Screen states

Every screen is designed in **all its states**, not only the happy one. Each screen has at least **Success, Loading, Empty and Error**, plus its own special states:

Screen	Special states
Nearby requests	Availability paused

Screen	Special states
Request details	Before accepting, accepted at a partner hospital (QR), accepted at a non-partner hospital (no QR), request closed
New request / Certificate	Uploading, checking, AI result, partner hospital, unclear → upload again, AI service down → review queue
Signup	Validation errors, submitting, account created
Confirm donation	Done, request expired
Hospital signup	Waiting for admin approval



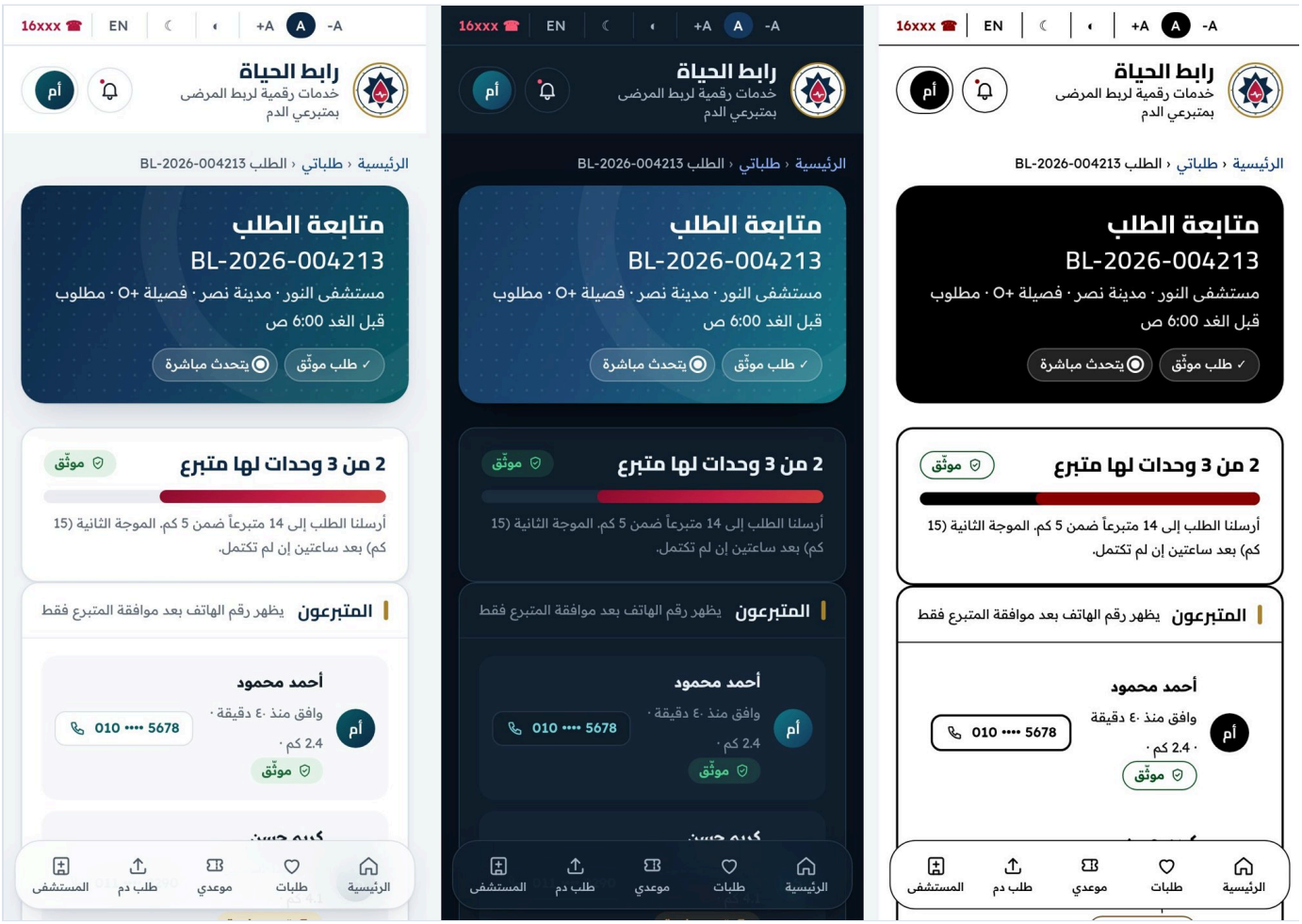
- **Loading** uses grey skeleton shapes in the same layout, so the page doesn't jump.
- **Empty** explains why and offers a next step ("We'll notify you when a compatible request appears within 5 km, then 15 km").
- **Error** says what happened in plain words, reassures ("nothing was lost"), and offers **Try again**.

8. Accessibility

Target: **WCAG 2.1 AA** and a Lighthouse accessibility score of at least 90.

Feature	How
Text size	A− / A / A+ buttons in the utility bar on every page. Layout uses `rem`, so everything scales.
High contrast	A switch that turns the page black-on-white with thick borders
Dark mode	A switch in the utility bar
Contrast	All text colours pass AA (4.5 : 1). Gold is used only for decoration, never for small text.
Keyboard	Everything works with Tab, Enter and Space. A "skip to content" link comes first.
Focus	A thick yellow focus ring with a navy halo, visible on any background
Screen readers	Real headings and landmarks, labels on every field, `aria-live` for upload progress and live request updates, icons hidden when a text label exists
Forms	Labels above fields, required fields marked with text, errors written next to the field and summarised at the top

Feature	How
Touch	Tap targets at least 44 px
Motion	Animations turn off when the phone asks for reduced motion



Arabic and right-to-left

- The whole page is `dir="rtl"` and `lang="ar"`.
- Layout uses start/end, never left/right, so it mirrors correctly.
- Phone numbers, reference codes (BL-2026-004213) and blood types (O+) are kept left-to-right inside Arabic text, so they never come out scrambled.
- Dates and times use Egyptian Arabic formatting and Cairo time.

9. Responsive design

The design is **mobile first**: most requesters and donors will use a phone.

Width	Layout
360–639 px (phones)	One column. Bottom tab bar. Tables turn into stacked cards. Side menus move above the content.
640–767 px	Header actions wrap below the logo
768 px and up (tablets)	Tables stay as tables. Two columns where useful.

Width	Layout
1024 px and up (computers)	Full main menu, side menu next to the content, up to 4 cards in a row

Every screen is designed for 1280, 768, 390 and 360 px wide.

10. Content and tone

- **Plain Arabic**, short sentences, no medical or legal jargon.
- **Say what happens next.** "We sent the paper to Al Nour Hospital. It usually confirms within 2 hours."
- **Never claim medical certainty.** "Paper looks valid", not "Paper is valid". "The hospital decides".
- **Repeat the 3 rules** where they matter: meet only inside the hospital, no money ever, the hospital makes the medical decision.
- **Privacy is visible.** "We won't share your phone number until you accept."
- **Reference numbers** on everything (requests, donors, appointments, reports), so people can talk to support about one exact item.

11. Usability testing plan

We will test the website with **5 people per round** (family and friends who match our users).

#	Task	Success when	Target
1	As a requester, send a request for 3 units of O+ at Al Nour Hospital	Request sent with a paper	Under 3 minutes
2	Find out how many donors have accepted your request	Finds "2 of 3" on the progress page	Under 30 seconds
3	As a donor, upload your certificate	Sees the badge result	Under 2 minutes
4	Accept a nearby request and find where to go in the hospital	Opens the QR ticket and the directions	Without help
5	Turn on high contrast and make the text bigger	Uses the utility bar	Without help

We watch, note where people hesitate, and fix the top problems before the next round. Results go into the Phase 6 test report.

12. From design to code

- **Design tokens in code:** the colours, fonts, sizes, radii and shadows above are defined once in `src/index.css` (Tailwind CSS v4), so every page uses exactly the same values and dark mode and high contrast work everywhere.
- **Shared components:** the components in section 6 are built once and reused by every page, with their variants and accessibility built in.
- **Every state for every screen:** each page is built with its loading, empty, error and success states, not only the happy one.

- **Checklist before merging:** every page is checked for keyboard use, focus, labels, text size, contrast, and the 390 and 360 px widths before it is merged.
- **Built with:** React, Vite and Tailwind CSS v4, in JavaScript (see the README).